

The Thermostat Closes as Structure: a fixed-point form for Λ with one named obstacle

Lyra (write-up), from the certified Lane Λ shot: Elie (assembly) + Grace (counts/cites co-sign), toy 5501, 6/6, quarantine-grep clean. 2026-08-26 (clock-verified at write). Certified content only; canonical scopes; no number is evaluated anywhere in this paper, and none was in the shot. Tier, on its face: CONDITIONAL — capped per K1057's lineage until discharge (see Section 4). Cell: External/cosmology.

For everyone (the five-minute version)

A thermostat keeps a house at one temperature not by making the furnace gentle but by *absorbing* what the furnace throws at it, cycle after cycle. The claim of this paper's form is that the cosmological constant works the same way: the boundary of the universe's geometry absorbs the enormous vacuum-energy “blast” each cycle, and Λ — the famously tiny leftover — is one cycle's un-absorbed mismatch, not the whole furnace. That is why it is small: not a miraculous cancellation of huge numbers, but the difference between a furnace and a draft under the door. What this paper does NOT do: produce the number. One ingredient — how much slips through per cycle — is not yet forced by the geometry. We name it, we do not guess it.

1. The fixed-point equation (the structure that closed)

The balance is written entirely from banked objects: the interior is held constant because the boundary absorbs the heat side's per-cycle action — absorption one-way (T2543, permanence) and monotone (T2546, the mean-side input) — and the residual is the un-absorbed mismatch per horizon cycle. The loop closes into a fixed point at $\tau_{\max} = 1/\Lambda$, the IR end of the ladder_heat dark-energy rung: Λ sets the horizon cycle, and the cycle's mismatch sets Λ . Self-consistency, not input.

The UV catastrophe appears exactly where it should and is absorbed exactly where the equation says: the a_0 -at-tick-cutoff term ($a_0 = 225$, F63) is what the boundary deformation eats — that absorption IS the thermostat holding the interior constant. The response coefficient the equation demands is $a_1 = -1875$ (F63, certified by alias sweep); it enters as “curvature change per unit absorbed action” because the equation's structure requires a response coefficient — the pre-committed fork resolved there, and no discrete-step demand ever fired.

2. The form (the deliverable)

$$\Lambda / \Lambda_P = (t_K / t_P)^{(2p / (2-p))}$$

with t_K the Koons tick, load-bearing and target-innocent by construction: $t_K = t_{\text{Planck}} \cdot \alpha^2(C_2^2)$ (T2405), derived as the substrate clock in May — three months before this lane existed. The suppression direction is structural: the residual is one cycle's mismatch, not a bulk sum. The suppression size waits on p (Section 3).

Counts, printed: t_K/t_P banked-valued (T2405) $\cdot a_0, a_1$ banked-valued (F63) $\cdot$ closure banked-structured (ladder_heat DE rung) $\cdot p$ FREE. free = 1, targets = 1 — the forced-exponent condition (free = 0) fails, so the evaluation landing could not fire and no evaluation step exists.

3. The one named obstacle

p = how many ticks' worth of action per horizon cycle escape absorption — equivalently, the 5D→4D reduction of the residual. It is not banked, it was not fit, and the shot did not guess it. The domain requirement inherits with the form (Grace's verification flag, landed at source): the exponent $2p/(2-p)$ is monotone on $p \in [0, 2)$ with a pole at $p = 2$ and a sign flip beyond — so a forced p must land in $(0, 2)$ for the form to suppress at all; a p outside that domain falsifies the form rather than resigning it. A future lane that forces p from D_{IV}^5 structure inherits this form with free = 0 — and this domain requirement with it — and computes the exponent blind under a fresh prereg. The 280-class quarantine transfers to that lane in advance: the coincidence-class number is target-not-evidence there exactly as it was here — never an input, never a target named in the derivation chain, evaluation last.

4. What passed, what caps, what is not claimed

Both controls passed. Must-catch: the balance's commits are clocked (T2405 is a clock, not a rate) and one-way (T2543); clocked one-way counting is sub-Poisson by structure — the suppressed-variance fact (T2571) is reproduced without being fed. Must-reject: allowing un-commit breaks the instrument both ways at once — equilibrium noise returns AND the fixed point fails to close. A structure that survives only when its physical input is true is evidence the input is doing work.

The cap, on its face: the ladder_heat chain rides K1057's channel-separation step, a CONDITIONAL pass (conclusion correct; the dimensional argument still owes the symmetry step for the dim-4 cross-term). Until that is discharged, everything here is CONDITIONAL — stated at both quotable sites, per the house rule that caps travel with quotations.

Not claimed: a value of Λ ; a value of p ; any contact with the measured number (the quarantine was never touched — with no evaluation, no measured value entered anything); any statement about sibling lanes beyond their own canonical forms.

— *Lyra, write-up of Elie + Grace's certified shot. The week's pattern holds here in its best form: the structure closed, the missing power got a name instead of a fit, and the form now waits — with its quarantine pre-transferred — for the lane that can force p .*